

Project Title
**(HEART DISEASE PREDICTION USING RIDGE- AND LASSO-
REGULARIZED LOGISTIC REGRESSION)**

Project Report submitted for the partial fulfilment of grade for the course

Machine Learning (25SC2107E)
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ABSTRACT

Heart disease is one of the leading causes of death worldwide, making early prediction and risk assessment important for timely medical intervention. This project focuses on developing a machine learning-based heart disease prediction system using **Logistic Regression** enhanced with **Ridge and Lasso regularization**. The model analyzes relevant patient attributes such as age, blood pressure, cholesterol levels, heart rate, and other clinical factors to predict the likelihood of heart disease.

Ridge regularization helps reduce overfitting by controlling the magnitude of model coefficients, while Lasso regularization performs feature selection by reducing less significant coefficients toward zero. By comparing these regularized models with standard Logistic Regression, the project aims to identify an effective and interpretable approach for heart disease prediction. The proposed system can assist in early risk assessment and support healthcare professionals in making informed decisions, while serving as a predictive aid rather than a replacement for clinical diagnosis.

Signature of the Guide